SUSHANTH PAATNAIK

INVENTOR · DEEP-TECH · FOUNDER

Engineering Matter. Engineering Capital. Engineering Scale.

From a maker's bench to global impact. This journey is about engineering breakthrough materials and ventures that build a more sustainable, abundant and equitable world.

ABOUT

SUSHANTH PAATNAIK

INVENTOR · DEEP-TECH FOUNDER · INDUSTRY BUILDER

Building industrial systems that outlive the inventor.

Sushanth Paatnaik is an inventor, deep-tech entrepreneur and industry builder focused on transforming breakthrough science into globally scalable industrial solutions. His work spans advanced materials, nanotechnology, clean energy and sustainable manufacturing, with multiple innovations recognized nationally and internationally.

PROFILE	ORIGIN	WORK BASE	EDUCATION
Inventor	Bhubaneswar, Odisha	Ahmedabad	B.Sc. @ IISER Bhopal
Deep-Tech Founder		New Delhi	B.E. (EC) @ OCT Bhopal
Industrial Strategist		Bhubaneswar	
Speaker			

FOCUS AREAS

- Advanced Materials
- Nanotechnology & Graphene
- Industrial Innovation
- Sustainable Manufacturing

> *"Engineering intelligent materials and scalable technologies that create lasting industrial impact."*

INVENTING · BUILDING · IMPACTING

ORIGIN

Every idea has a birthplace.

Childhood curiosity in

a small town

First workshop

at age 14

Breath-powered wheelchair

that started it all

PHILOSOPHY

The future is not imagined. It is engineered quietly.

EARLY WORKS

The teenage innovations that earned six Presidential Awards.

Between 2008 and 2013 — still in his teens — Sushanth designed and built
six original devices, each tackling a real human problem with hardware that worked.

BREATH-POWERED WHEELCHAIR

2008

Developed India's first wheelchair operated using human breath. Recognized by the President of India and became the foundation of my innovation journey.

ACCIDENT-PROOF RETROFIT KIT

2011

A retrofittable safety kit designed to transform any existing vehicle into a safer machine — without redesign of the host platform.

SHE WATCH

2012

A discreet, smart-watch form factor wearable engineered to alert and locate — designed long before the category became mainstream.

SOL-KA

2012

Ambient-light charging power bank that harvests indoor light to keep devices topped up — a quietly self-sufficient piece of everyday hardware.

SUPER SENSE TECHNOLOGY

2013

Gesture-controlled computing enabling computers to be operated through hand gestures alone — exploring a natural human-machine interface.

HIGH-EFFICIENCY POWER KIT

2008

A high-efficiency, low-cost power generation kit conceived to bring reliable electricity to remote rural communities.

“ *Every idea was a small step towards solving a real problem.*
These early innovations shaped the path ahead. ”

Sushanth Paatnaik

SUSHANTHPAATNAIK.COM

JOURNEY

A TIMELINE OF INNOVATION, IMPACT & INDUSTRY CREATION

From a young inventor to a deep-tech entrepreneur,
building technologies and companies that solve real-world problems.

2008
FIRST PRESIDENTIAL AWARD

2010
MIT TR35

2010
INK FELLOW

2012
TED SPEAKER

2012
NASA AWARDEE

2013
SIX PRESIDENTIAL AWARDS

2015
INTHINKS FOUNDED

2016
ZZEN-X INVERTER LED

2020
CAPATTERY

2024
SPI INDUSTRIES

2025
MONOATOM LABS

2025
GRAFILLIUM

TODAY
GLOBAL DEEP-TECH ECOSYSTEM

> *“Every idea was a small step towards solving a real problem.
> These early innovations shaped the path ahead.”*

Sushanth Paatnaik

SUSHANTHPAATNAIK.COM

HONORS & RECOGNITIONS

PRESIDENT OF INDIA

AWARDS X 6

Honored by the President of India for six consecutive years for outstanding innovation and leadership.

NASA

AWARDEE

Recognized by NASA for contributions in innovation and technology.

MIT TR35

MIT TECHNOLOGY REVIEW 35 UNDER 35

Recognized among the world's top young innovators under 35.

TED

SPEAKER

Invited TED Speaker sharing ideas that inspire and create impact.

intel.

INTEL INNOVATOR

Recognized by Intel for driving breakthrough technologies.

INK

INK FELLOW

Honored by India Innovation Network for exceptional innovation and impact.

NIF

NIF AWARDEE

Recognized by the National Innovation Foundation for grassroots innovation.

HALL OF FAME

RECOGNITIONS · AWARDS · HONORS

A journey of innovation, impact and inspiring a better tomorrow.

2009
National Recognition

2010
Youth Leadership

2013
Innovation Champion

◆ **PRESIDENT OF INDIA**

SIX-TIME NATIONAL AWARDEE

2008 · 2009 · 2010 · 2011 · 2012 · 2013

India's Highest National Recognition
for Young Innovators

NASA Award
2011

MIT TR35 Innovator
2011

Innovation Showcase
2010

Fab-10 Barcelona
2013

Fab-11 MIT Boston
2014

INK Fellow
2010

Intel IRIS Award
2010

Young Achiever
2013

STPI Chunauti
2021

Elecrama Recognition
2024

CEO Clubs Leadership
2025

BRICS Global Delegate
2023

FICCI Bharat Summit
2024

Manufacturing Leadership
2023

Industry & Innovation Excellence
2025

Mobility Innovation
2019

Social Impact Recognition
2018

Youth Icon Award
2016

Entrepreneurship Excellence
2022

National Innovation Recognition
2017

A LEGACY OF INNOVATION

2008	**2010**	**2012**	**2013**	**2016**	**2018**	**2021**	**2025**
Journey Begins	Early Recognitions	National Honors	Global Milestones	Expanding Impact	Inspiring Change	Building The Future	Continuing The Mission

INNOVATOR · ENTREPRENEUR · SPEAKER
ADVISOR · MENTOR · DEEP-TECH BUILDER

Sushanth Paatnaik
SUSHANTHPAATNAIK.COM

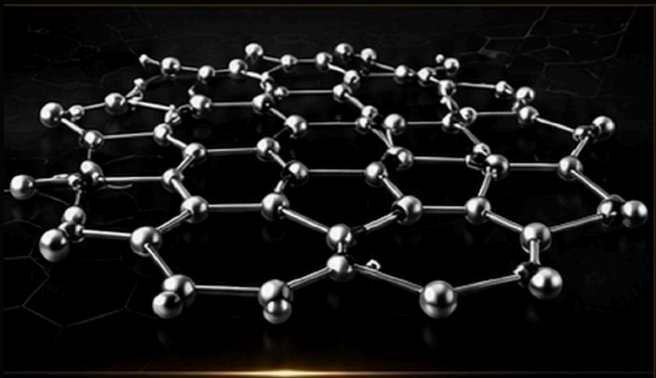
BUILDING INTELLIGENT MATTER.
TRANSFORMING INDUSTRIES.
IMPACTING BILLIONS.

BY THE NUMBERS

A track record of innovation, impact
and global recognition.

15+ — YEARS	10+ — INDUSTRIAL TECHNOLOGIES
6 — PRESIDENTIAL AWARDS	5 — VENTURES
50+ — MEDIA FEATURES	100+ — INDUSTRY COLLABORATIONS

ENGINEERING INTELLIGENT MATTER

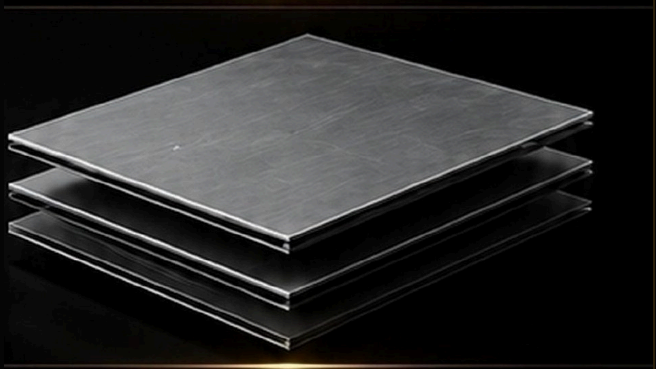


GRAPHENE

The foundation of next-generation materials. Stronger, lighter, smarter by design.

ADDITIVES

Nano-engineered additives that enhance performance, efficiency, and sustainability.

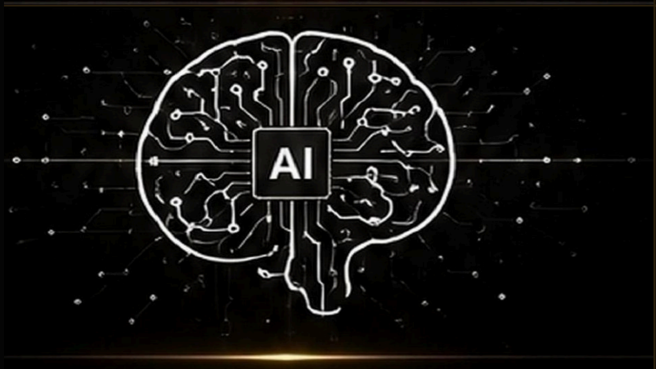


COATINGS

Advanced functional coatings for protection, performance, and precision.

COMPOSITES

High-performance composites engineered for tomorrow's challenges.



AI

Intelligent systems and data-driven innovation powering the future of materials.

23

INNOVATIONS ENGINEERED FOR IMPACT

- ADVANCED MATERIALS
- CLEAN ENERGY
- INDUSTRIAL SOLUTIONS

“

Engineering intelligent materials and scalable technologies that create lasting industrial impact.

23
INNOVATIONS
Across Diverse Industries

3
DEVELOPMENT STAGES
Commercial · Pilot · R&D

SUSTAINABLE IMPACT
Building technologies for a better, stronger and cleaner planet

STAGE I – COMMERCIAL / FIELD DEPLOYED (1–6)

1 GRAPHACRETE
Graphene Concrete Admixture

CONSTRUCTION

- ↑ 30–45% Strength
- ↓ 15–25% Cement Use
- ↓ 20–30% CO_2 Emissions

2 GRAFFISOL
Graphene Solar Coating

SOLAR ENERGY

- ↑ 5–12% Solar Efficiency
- ↓ 10–18°C Surface Temperature
- ↑ 20% Panel Life

3 CERAPHENE
Graphene Ceramic Coating (Car Body)

CAR PROTECTION

- ↑ 3x Hardness
- ↑ 60% Wear
- ↓ 5x Corrosion Resistance

4 HD-G-PE
Graphene HDPE Masterbatch

POLYMERS

- ↑ 30–40% Tensile Strength
- ↓ 15% Material Usage
- ↑ 2x Service Life

5 GRAPHENODES
Graphene Electrode Materials

BATTERIES

- ↑ 25–40% Energy Density
- ↑ 3x Charging Speed
- ↑ 2x Cycle Life

6 IGNITRON D
Diesel Combustion Optimization

MOBILITY

- ↑ 8–15% Fuel Efficiency
- ↓ 35–60% Smoke
- ↓ 20–30% Emissions (NOx/CO)

STAGE II PLANT & FIELD PILOT (7–16)

7 COALORIX
Coal Combustion Optimization

THERMAL POWER
- ↑ 5–10% Coal Consumption
- ↓ 3–6% Boiler Efficiency
- ↓ 15–30% Fly Ash

8 AQUAMAX
High Voltage, Low Current Water Recovery System

WATER
- ↑ 30–50% Water Recovery
- ↑ 15–20% Water Reuse
- ↑ 15–20% Cooling Efficiency

9 IGNITRON P
Petrol Combustion Optimization

MOBILITY
- ↑ 6–12% Mileage
- ↓ 20–30% Emissions

10 LUBRITRON
Nano Engine Oil Additive

LUBRICANTS
- ↓ 40% Engine Wear
- ↑ 2x Oil Life
- ↑ 3–6% Fuel Economy

11 RUSTENE
Anti-Corrosion Coating

COATINGS
- ↑ 5x Corrosion Protection
- ↑ 3x Coating Life

12 GRYOGEN
Hydrogen Separation Membrane

H_2 HYDROGEN
- ↑ 99.9%+ Hydrogen Purity
- ↓ 30% Separation Energy

13 MARIPHENE
Desalination Membrane

WATER
- ↓ 20–35% Energy Consumption
- ↑ 30% Water Recovery

14 AEROPHENTER
Atmospheric Water Harvesting

WATER

Produces 5–25 L/day of Water per Unit*
(Climate Dependent)

15 FIBRASHENE
Reinforced Graphene Fibres

COMPOSITES
- ↑ 40–60% Tensile Strength
- ↓ 25% Weight

16 VOLTAPHENE
Graphene Battery Cell

ENERGY STORAGE
- ↑ 2x Lifespan
- ↑ 3x Fast Charging Capability
- ↑ 30% Energy Density

STAGE III R&D / BENCH (17–23)

17 ARMOPHENE
Graphene Armour

DEFENCE
- ↑ 30–40% Weight Enhanced Ballistic Resistance

18 HYDROCELL
Fuel Cell Technology

H_2 HYDROGEN
- ↑ 15–20% Fuel Cell Efficiency
- ↓ Catalyst Usage

19 BITUMAX
Bitumen Additive

INFRASTRUCTURE
- ↑ 2x Pavement Life
- ↓ 40–60% Rutting & Cracking

20 PYRONEX
Fire & Thermal Protective Additive

COATINGS
- ↑ Fire Resistance up to 1200°C
- ↑ 30% Thermal Insulation

21 GRAPHYRE
Graphene Tyres

MOBILITY
- ↑ 40–60% Tyre Life
- ↓ 15% Rolling Resistance

22 GRAPHOSITE
Structural Graphene Composites

COMPOSITES
- ↑ 35–50% Structural Strength
- ↓ 25–40% Weight

23 THERMAPHENE
Temperature-Regulating Textile

SMART MATERIALS
- ↑ 5–8°C Passive Temperature Reduction
- ↑ Thermal Comfort

SUSTAINABLE
Solutions that reduce emissions, conserve resources and protect our planet.

SCALABLE
Technologies engineered for scale and real-world industrial adoption.

IMPACTFUL
Delivering measurable performance, efficiency and value.

GLOBAL VISION
Innovations for industries and communities worldwide.

Portable RO with Reject-Water Recovery

Conventional portable RO discards three litres for every one it purifies. This proposal pairs a closed-circuit recirculation loop with a graphene-oxide recovery cartridge to reclaim a significant share of it — without giving up the compact footprint.

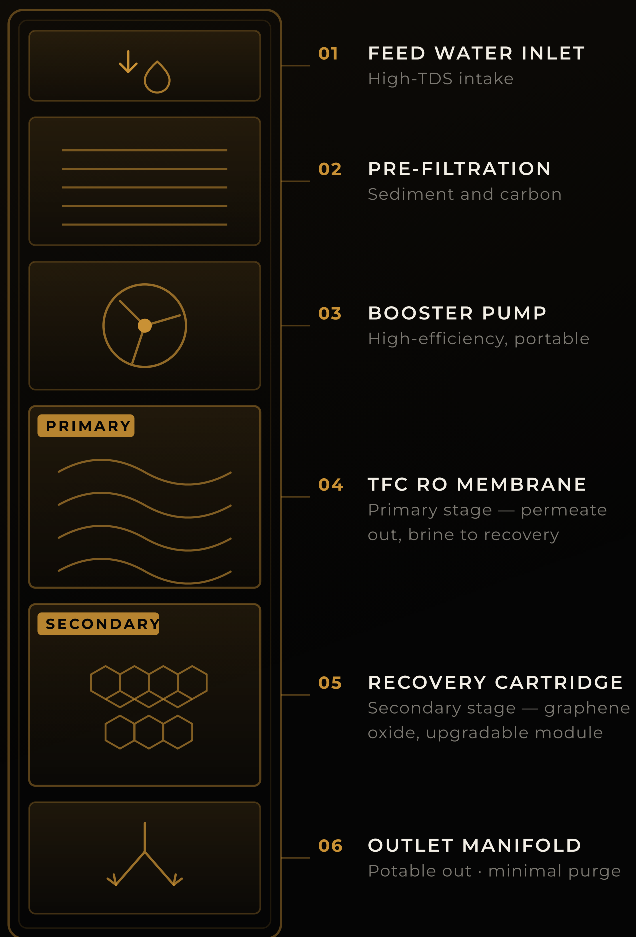
THE PROBLEM

RO forces water through a semi-permeable membrane with pore sizes near 0.0001 microns. It is highly effective at reducing Total Dissolved Solids and removing heavy metals, but standard residential and portable units run a 3:1 waste ratio — 75% of intake is discarded to stop mineral crystallisation scaling the membrane.

THE ARCHITECTURE

Primary stage. A high-efficiency booster pump and a Thin Film Composite membrane produce potable water, separating salts and hardness minerals into a concentrated brine stream.

Secondary stage. Rather than routing brine to a drain, it is diverted into a recovery attachment built on advanced materials and modified flow dynamics — because standard TFC membranes cannot survive the osmotic pressure or the rapid fouling that concentrated brine causes.



UNIFIED PORTABLE HOUSING · TWO-STAGE SCHEMATIC

> A next-generation portable purifier that processes its own reject water — significantly increasing the overall recovery rate while maintaining a compact footprint.
>
> PROJECT PROPOSAL · EXECUTIVE SUMMARY

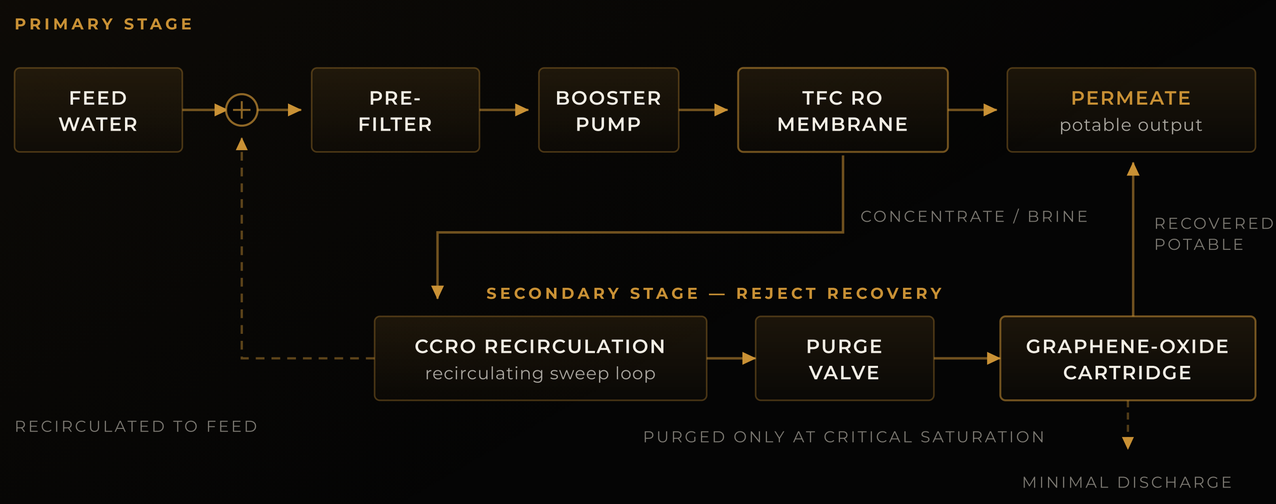
3:1 — CONVENTIONAL WASTE RATIO

0.0001 — MICRON MEMBRANE PORE SIZE

75% — OF INTAKE DISCARDED BY STANDARD UNITS

HOW IT RECOVERS

Closed-circuit recirculation, graphene-oxide recovery.



SYSTEM ARCHITECTURE · TWO-STAGE RECOVERY FLOW

ECONOMIC FEASIBILITY FOR PORTABLE UNITS

TECHNIQUE	CAPEX / OPEX	COMMERCIALISATION ASSESSMENT
Graphene Oxide (GO) Membranes	Moderate–High CAPEX Low OPEX	High initial setup cost — scaling uniform nanomaterial manufacturing is hard. Best portable performance: low pressure requirement, high fouling resistance. Ideal for premium positioning.
Miniaturised CCRO	Low CAPEX Moderate OPEX	Highly economical. Uses existing pump architecture plus smart valves and sensors. Excellent candidate for immediate commercialisation.
Electrodialysis Reversal (EDR)	High CAPEX Low OPEX	Miniature electrodes and ion-exchange membranes remain prohibitive for consumer portables, but viable for specialised military or emergency-relief systems.
Thermal Evaporation (ZLD)	Highly uneconomical	The energy required to boil brine is massive. Completely unsuited to portable, battery-operated or low-power applications.

STRATEGIC RECOMMENDATION

A hybrid approach. The primary stage runs a standard TFC membrane with a CCRO recirculation loop to cut waste economically; the secondary reject attachment becomes an upgradable graphene-oxide cartridge, targeting the premium segment where higher CAPEX is justified by maximum water recovery in off-grid scenarios.

INDUSTRIAL TRANSLATION

TRANSFORMING INNOVATION INTO REAL-WORLD IMPACT

SOLAR

Advanced coatings for higher efficiency and durability.

ENERGY

Materials and technologies driving cleaner, smarter energy solutions.

AUTOMOTIVE

High-performance materials for mobility and efficiency.

INFRASTRUCTURE

Stronger, smarter solutions for resilient and sustainable infrastructure.

DEFENSE

Advanced composites for protection, performance, and mission criticality.

CONSTRUCTION

Innovative materials for stronger, longer-lasting and sustainable builds.

VENTURES

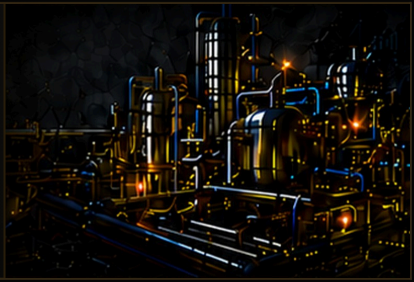
SPI INDUSTRIES
R&D INDUSTRIAL SOLUTIONS

SPI INDUSTRIES

R&D Industrial Solutions

FOUNDER

Driving industrial innovation through advanced R&D and engineering solutions.



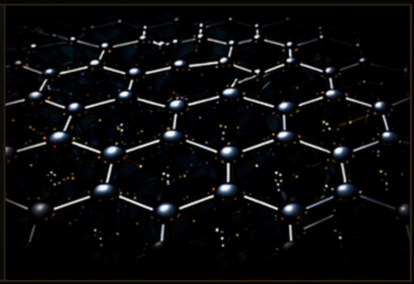
MONOATOM
LABS

MONOATOM LABS

Graphene at Scale

CO-FOUNDER & CEO

Pioneering graphene production at scale and developing innovative applications.



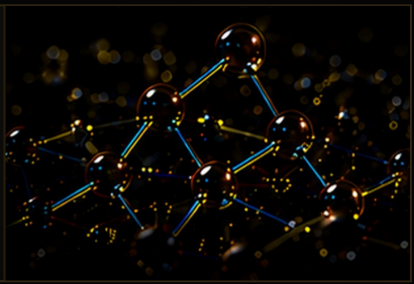
GRAFFILLIUM
DEEP TECH
INNOVATIONS

GRAFFILIUM

Eco-Nano Additives

CO-FOUNDER & CIO

Developing eco-nano additive solutions for industrial and energy applications.



INTHINKS
INNOVATION STUDIO

INTHINKS

Innovation Studio

FOUNDER

Deep-tech research and product development for real-world challenges.

STARUNICO
INVESTMENT

STARUNICO CAPITAL

Deep-Tech Investment

FOUNDER

Investing in breakthrough deep-tech startups building a sustainable and intelligent future.

MAGPPIE

MAGPPIE

Stone Wellness Kitchens

CHIEF INNOVATION OFFICER (CIO)

Pioneering the world's first 100% stone-built modular kitchen solutions.

◆ COUNSEL ACROSS DIVERSE INDUSTRIES. ◆

A short ledger of the houses I quietly advise — one mark per layer of the industrial network.

VPRPL | WEHEAR | Tileopedia | EUROGRES | GREENOMERS GREEN POLYMERS

NEWS & MEDIA

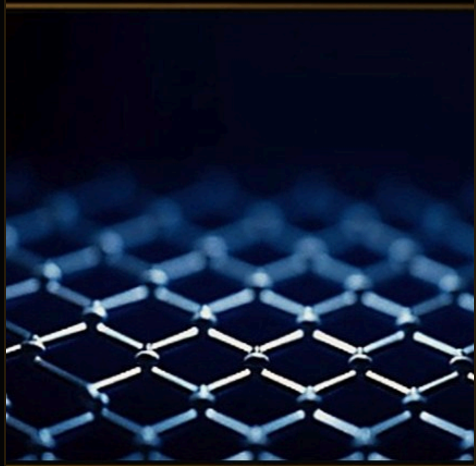
A LEGACY DOCUMENTED BY THE WORLD'S LEADING PUBLICATIONS.

From national headlines to global platforms, the journey of innovation, impact and entrepreneurship continues to inspire.

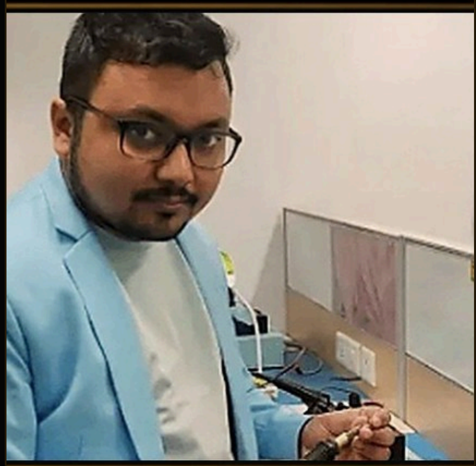
16+	50+	COVERAGE ACROSS
YEARS OF MEDIA PRESENCE	NATIONAL & INTERNATIONAL FEATURES	NATIONAL & GLOBAL PLATFORMS

FEATURED COVERAGE

Six times President awardee profile
The Global Indian | 2013



GraphIN 2026 graphene coverage
Rediff / PTI | 2026



Battery charging innovation
Business Standard | 2024

Serial entrepreneur profile
India Today | 2016

Patent run and awards
Times of India | 2012

Capattery graphene energy storage
YourStory | 2023

FEATURED IN

India Today | The Times of India | Business Standard | ThePrint | Deccan Chronicle | The Telegraph online

The Global Indian | MIT Technology Review | TED India | National Innovation Foundation | Governance now | rediff.com PTI

YourStory | Product Nation | WeRIndia | The New Indian Express | INK

“From local headlines to global platforms,
the journey of innovation continues to inspire.”

SPEAKING ENGAGEMENTS

From thought leadership talks to global conferences, sharing insights on innovation, deep-tech, and building a sustainable future.

TED SPEAKER

Delivered a thought-provoking talk on innovation and the future of technology at a TEDx event.

CEO CLUBS & LEADERSHIP FORUMS

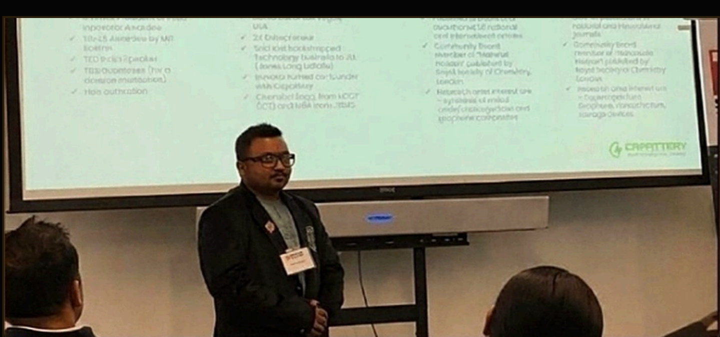
Engaged with industry leaders and CXOs to discuss strategy, growth, and the future of deep-tech ventures.

UNIVERSITIES & INSTITUTES

Invited by leading universities to speak on entrepreneurship, deep-tech innovation, and building impactful solutions.

GLOBAL CONFERENCES

Participated in global platforms to share insights on emerging technologies and sustainable innovation.



SILICON VALLEY & BEYOND

Connected with innovators and investors across Silicon Valley and beyond to explore the future of deep-tech.

VOICES

ENDORSEMENTS & TESTIMONIALS

What people say about the work.

On-record testimonials from innovation foundations, global press, industry leaders, and international stages.

06	20+	40+	GLOBAL
PRESIDENTIAL AWARDS	INSTITUTIONAL VOICES	EDITORIAL FEATURES	STAGES & OUTLETS

National Innovation Foundation
Accelerating Inclusive Innovations

“ Sushant is an amazing innovator and always innovates with high social impact.

– Anil K. Gupta
Professor, IIM-Ahmedabad
Vice-Chair

MIT Technology Review MAGAZINE

“ 'A life-changing innovation 'Enabler' for the disabled — highly appreciable.'

– Mr. Srinivas
Senior Journalist

Deloitte.

“ The best AI live demo ever seen so far. This demo really made my day.

– P. R. Ramesh
Chairman

इंडियनऑयल IndianOil

“ Very effective and innovative solution for underground pipelines.

– Mr. Joseph
Eastern Zone

YOUR STORY

“ *A very inspiring entrepreneur and great social-revolutionary products.*

– Shradha Sharma
Founder & CEO ”

COLLABORATION

INDUSTRY Co-creating impact at scale	**INVESTMENT** Funding deep-tech moonshots	**RESEARCH** Pushing the boundaries of science	**GOVERNMENT** Building systems for the future

Let's build the future together.

sushanthpaatnaik.com

Building industrial systems that outlive the inventor.

SUSHANTH PAATNAIK

sushanthpaatnaik.com

info@sushanthpaatnaik.com

Ahmedabad | New Delhi | Bhubaneswar